



Acsse CSC3B 2023 SAO1 MAIN

Computer Science (University of Johannesburg)



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FACULTY OF SCIENCE

ACADEMY OF COMPUTER SCIENCE AND SOFTWARE ENGINEERING

MODULE	COMPUTER SCIENCE 3B CSC03B3
CAMPUS	AUCKLAND PARK CAMPUS (APK)
SUMMATIVE ASSESSMENT OPPORTUNITY 1	SAO1

DATE: 2023-08-31	SESSION: 14h00 - 16h00
ASSESSOR(S):	DR J. DU TOIT MR. A. MAGANLAL
MODERATOR:	INTERNAL: MR. S. SITHUNGU
DURATION: 120 MINUTES	MARKS: 100

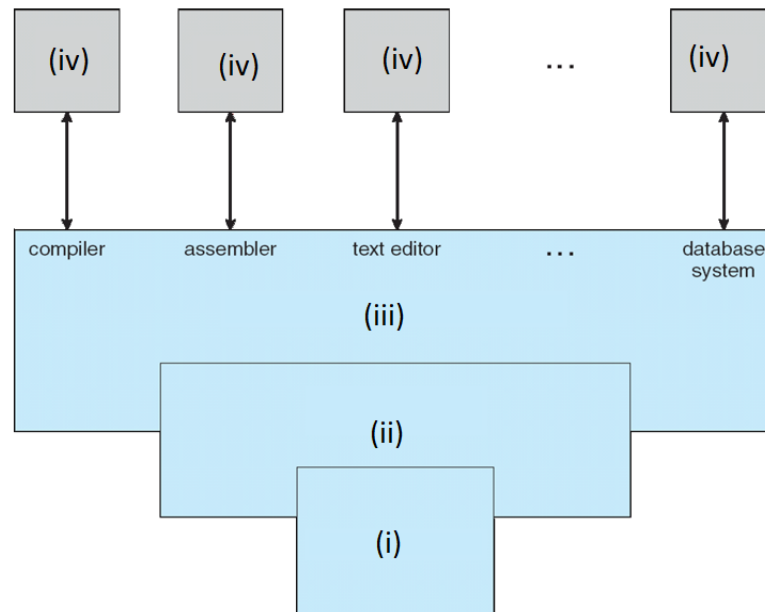
Please read the following instructions carefully:

1. You are bound by all university regulations including, but not limited, to assessment, plagiarism, and ethical conduct.
2. You may not leave the venue in the first 30 minutes of the paper.
3. You may not leave the venue in the last 15 minutes of the paper.
4. Answer **all** the questions.
5. Answer questions in order.
6. Answer only in the answer sheets provided.
7. The use of calculators is *not* permitted.
8. Write ***cleanly and legibly***.
9. This paper contains **5** questions.
10. This paper consists of **3** pages excluding the cover page.

QUESTION 1: Operating Systems - General

(a) **Identify** components i - iv depicted in the diagram below.

[04]



(b) **Name and briefly describe** the *hardware components* involved when a **file is saved**.

[08]

(c) **State** which of the following *memory* is either **volatile** or **nonvolatile**.

i. Main Memory

[01]

ii. Hard disk

[01]

iii. USB Memory stick

[01]

iv. Cache Memory

[01]

(d) **Briefly describe** the *differences* between **symmetric multiprocessing** and **asymmetric multiprocessing**.

[02]

(e) **Discuss** the *role* each of the four *storage-devices* play when a **program is executed**. The four storage-devices are part of the *storage-device hierarchy*.

[08]

(f) One of the activities performed by operating systems is *memory management*. **Briefly describe** two **memory management activities**.

[04]

Total: 30

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QUESTION 2: Operating-System Structures

- (a) **Describe** the *advantage* for each of the following:
- i. **Command line interface** [02]
 - ii. **Graphical user interface** [02]
- (b) With the *aid of a diagram*, **describe** how a **system call** is *handled* from the **user application** all the way to the **operating system** and back. [08]
- (c) **Analyse** each of the following statements. Clearly state whether the statement is **TRUE** or **FALSE** and **motivate** your answer.
- i. **Operating systems** are *written* purely in **assembly language**. [02]
 - ii. An operating system structure that uses loadable kernel modules is a form of a layered design. [02]
 - iii. System calls can pass parameters using a register. [02]
- (d) **Describe** how a two-step bootstrap starts the operating system. [02]

Total: 20

QUESTION 3: Processes

- (a) **Name and motivate** the **process state** for each of the following scenarios
- i. A word processor is loading a document. [02]
 - ii. A word processor has just finished loading a document. [02]
 - iii. A word processor is scanning a document for spelling mistakes. [02]
- (b) **List** four *parts* of a **process**. [04]
- (c) **Briefly** describe how a context switch occurs. [04]
- (d) **Describe** indirect inter process communication (IPC) and **explain** the main difference between direct and indirect (IPC) [05]
- (e) **Analyse** each of the following statements. Clearly state whether the statement is **TRUE** or **FALSE** and **motivate** your answer.
- i. The following instruction is an example of IPC using a pipe: **dir | more** [02]
 - ii. In Remote Procedure Calls, the matchmaker service returns the name of the server service. [02]
 - iii. All ports below 1024 is normally associated with non-standard services. [02]

Total: 25

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QUESTION 4: Number Systems

Show all calculation steps!

- (a) **Hexadecimal** and **octal** can be seen as shorthand for binary. **Solve** the following equation [05] which contains terms in multiple number systems and provide the answer in the **hexadecimal** (e.g. $q = 17_{16}$):

$$x_{16} = 00001110_2 + 53_8 + 4C_{16}$$

- (b) **Convert** the number below into *two's complement* binary representation. Use exactly 8- [05] bits to represent the number. Show all the steps in the calculation.

$$-80_{10}$$

Total: 10

QUESTION 5: 80x86 Assembly

Provide code for 80x86 MASM assembly function that is equivalent to the following C++ function:

```
1 void reducer(int *nums, int len, int s)
2 {
3     for (int i = 0; i < len; i++) // Iterate through array
4     {
5         s += nums[i]; // sum values
6     }
7 }
```

Only provide code for the function. No global variables are to be used.

Total: 15

~~ THE END ~~